PERFECTNESS IS CHECKED ON THE STALKS OF A PSEUDO-COHERENT MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 17913-17951, immediately following the statement at 17900-17911.

Editorial note. Four cross-chapter pointers (Cohomology, More on Algebra) are recorded in assembly.external_references.

Editorial note. The reduction to the affine case and the polynomial ring representation are displayed inside the span.

Allowed standard prerequisites (editorial).

- Tor amplitude on stalks. Cohomology, lemma cohomology-lemma-tor-amplitude-stalk, cited in the proof. Converts a local tor dimension condition into a stalkwise one.
- Localisation and tor amplitude. More on Algebra, lemma more-algebra-lemma-tor-amplitude-localization, cited in the proof. Moves tor amplitude along localisations of the ring map.
- Perfectness over a polynomial ring. More on Algebra, lemma more-algebra-lemma-perfect-over-polynomial-ring, cited at proof lines 17943-17946. Upgrades finite tor dimension over the localised polynomial ring to perfectness there.
- Checking perfectness on stalks of a finitely presented algebra. More on Algebra, lemma more-algebra-lemma-check-perfect-stalks, cited at proof lines 17948-17949. Concludes that the module is perfect, hence the ring map is perfect, from the stalkwise information.

Source reference. Original reference 09U9: focused-seeds/geometry/S9-morph-check-perfect-stalks/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 068W: focused-seeds/geometry/S9-morph-check-perfect-stalks/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 068X: focused-seeds/geometry/S9-morph-check-perfect-stalks/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B67: focused-seeds/geometry/S9-morph-check-perfect-stalks/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a pseudo-coherent morphism of schemes. The following are equivalent*

(1) *f is perfect,*
(2) *$\mathcal{O}_X$ locally has finite tor dimension as a sheaf of $f^{-1}\mathcal{O}_S$-modules, and*
(3) *for all $x \in X$ the ring $\mathcal{O}_{X,x}$ has finite tor dimension as an $\mathcal{O}_{S,f(x)}$-module.*

Proof. The problem is local on X and S. Hence we may assume that $X = \mathrm{Spec}(B)$, $S = \mathrm{Spec}(A)$ and f corresponds to a pseudo-coherent ring map $A \to B$.

If (1) holds, then B has finite tor dimension d as A-module. Then $B_{\mathfrak{q}}$ has tor dimension d as an $A_{\mathfrak{p}}$-module for all primes $\mathfrak{q} \subset B$ with $\mathfrak{p} = A \cap \mathfrak{q}$, see More on Algebra, Lemma 0B67. Then $\mathcal{O}_X$ has tor dimension d as a sheaf of $f^{-1}\mathcal{O}_S$-modules by Cohomology, Lemma 09U9. Thus (1) implies (2).

By Cohomology, Lemma 09U9 (2) implies (3).

Assume (3). We cannot use More on Algebra, Lemma 0B67 to conclude as we are not given that the tor dimension of $B_{\mathfrak{q}}$ over $A_{\mathfrak{p}}$ is bounded independent of $\mathfrak{q}$. Choose a presentation $A[x_1, \ldots, x_n] \to B$. Then B is pseudo-coherent as a $A[x_1, \ldots, x_n]$-module. Let $\mathfrak{q} \subset A[x_1, \ldots, x_n]$ be a prime ideal lying over $\mathfrak{p} \subset A$. Then either $B_{\mathfrak{q}}$ is zero or by assumption it has finite tor dimension as an $A_{\mathfrak{p}}$-module. Since the fibres of $A \to A[x_1, \ldots, x_n]$ have finite global dimension, we can apply More on Algebra, Lemma 068X to $A_{\mathfrak{p}} \to A[x_1, \ldots, x_n]_{\mathfrak{q}}$ to see that $B_{\mathfrak{q}}$ is a perfect $A[x_1, \ldots, x_n]_{\mathfrak{q}}$-module. Hence B is a perfect $A[x_1, \ldots, x_n]$-module by More on Algebra, Lemma 068W. Thus $A \to B$ is a perfect ring map by definition. □